CivilSuite — Help: formulas and references

Help version 1.0 (October 2026). This help describes what the app actually computes. Where a piece of information could not be verified, it is marked “I don't know”.

> **Disclaimer.** CivilSuite is a calculation and decision-support tool. Results must be checked by a responsible engineer against the standards in force and the National Annex of the project country. Mix designs are starting mixes to be validated by trial batches.

1. General principles

- **Calculations always in SI.** All values are stored and computed in SI units; the “Units” setting only changes input and display (screen and PDF). Calculation steps remain expressed in SI.
- **PDF language.** Chosen in Settings → PDF reports (default: device language), can be changed for each export from the preview. Numbers and dates follow the chosen language; text typed by the user is never translated.
- **Read-only.** A Eurocode calculation opened in US units can still be viewed (converted values) and exported to PDF, but cannot be edited until you switch back to SI or choose an ACI method.

2. SI units and US customary units

“US customary” means the units commonly used in the United States (in, ft, lb, psi, °F, US gallon). It is not the British imperial system: the imperial gallon is 4.546 09 L, the US gallon 3.785 411 784 L.

Why only ACI and ASTM methods in US units?

Eurocodes and EN standards are published in SI units only: their coefficients (for example $f_{bd} = 2.25\eta_1\eta_2 f_{ctd}$ or the C30/37 classes) are calibrated in MPa and mm. A “US version” of a Eurocode calculation would be a derivation with no normative standing. Conversely, ACI 318 and ASTM standards exist natively in US units, with their own constants: these are not rounded conversions of the SI values. Example (ACI 318-19, required average strength without data): +1000 psi, +1200 psi, then 1.10 f′c + 700 psi, whereas the SI edition gives +7.0 MPa, +8.3 MPa, then 1.10 f′c + 5.0 MPa; 8.3 MPa = 1204 psi ≠ 1200 psi. In US units the app therefore uses the native psi values.

Conversion factors (exact definitions)

Quantity	Definition
Length	1 in = 25.4 mm; 1 ft = 0.3048 m; 1 yd = 0.9144 m
Mass	1 lb = 0.453 592 37 kg; 1 short ton = 2000 lb
Force	1 lbf = 4.448 221 615 260 5 N; 1 kip = 1000 lbf
Stress	1 psi = 1 lbf/in² = 6 894.757… Pa
Volume	1 US gal = 3.785 411 784 L; 1 yd³ = 0.764 554 857 984 m³
Temperature	$°\mathrm{F} = 1.8 \times °\mathrm{C} + 32$

Source: NIST SP 811 (Appendix B); 1959 international yard and pound agreement.

Reinforcement: rebar mass, bar shapes and mesh stay in metric units in both systems (app design choice). Mesh lapping (Eurocode) is read-only in US units.

3. Concrete mix design

3.1 Eurocode / EN 206 (Bolomey method)

Target mean strength:

$$f_{cm} = f_{ck} + 8 \quad (\text{EN 1992-1-1 Table 3.1}) \quad \text{or} \quad f_{cm} = f_{ck} + 1.48\sigma$$

The 1.48 σ criterion is the EN 206 conformity criterion for continuous production. Cement/water ratio from Bolomey's law (G and A are empirical, to be calibrated on tests):

$$f_{cm} = G \cdot f_{ce} \cdot \left(\frac{C}{W} - 0.5 \right)$$

The retained w/c is the minimum of the “strength” w/c and the maximum w/c of the exposure class (EN 206 Annex F, recommended values; the National Annex prevails); binder content is raised to the exposure minimum where needed. As EN 206 gives no water-demand method, the initial water and air estimate comes from the ACI 211.1 tables. Bolomey reference curve and least-squares optimisation of the proportions:

$$P(d) = A + (100 - A)\sqrt{\frac{d}{D}} \quad ; \quad \min \sum_{\text{sieves}} \left(P_{\text{blend}} - P_{\text{ref}} \right)^2$$

3.2 ACI 211.1 / ACI 318

Required average strength f′cr (ACI 318-19 §26.4.3.1). With a known standard deviation s (SI edition):

$$f'_{cr} = \max\left(f'_c + 1.34s;\ f'_c + 2.33s - 3.5\right) \quad \text{if f'c} \le 35\ \text{MPa}$$

$$f'_{cr} = \max\left(f'_c + 1.34s;\ 0.90f'_c + 2.33s\right) \quad \text{if f'c} > 35\ \text{MPa}$$

No production data	SI edition	US edition (psi)
Low f′c	f′c < 21 MPa: f′c + 7.0	f′c < 3000: f′c + 1000
Medium f′c	21–35 MPa: f′c + 8.3	3000–5000: f′c + 1200
High f′c	> 35 MPa: 1.10 f′c + 5.0	> 5000: 1.10 f′c + 700

In US units with a standard deviation: max(f′c + 1.34 s; f′c + 2.33 s − 500) up to 5000 psi, max(f′c + 1.34 s; 0.90 f′c + 2.33 s) above. Then: water and air (ACI 211.1 Table 6.3.3), w/c (Table 6.3.4(a), interpolated), coarse aggregate (Table 6.3.6, from the sand fineness modulus), sand by absolute volumes, moisture corrections. Minimum f′c for exposure follows ACI 318-19 Table 19.3.2.1 (native psi values in US units: 2500, 3500, 4000, 4500, 5000 psi depending on the class).

4. Crack diagnosis

Photo measurement. The user draws a transect across the crack and a scale (line of known length or QR code of known side). The width is the full width at half maximum (FWHM) of the intensity dip, with a ± 1 pixel uncertainty. The reliability thresholds (3 px, 15 % contrast) are the app's own, not normative. A crack-width gauge remains the reference.

Width limits. EN 1992-1-1 Table 7.1N (recommended w_{max}, quasi-permanent combination): 0.4 mm (X0, XC1), 0.3 mm (XC2–XC4, XD, XS). ACI 224R-01 Table 4.1 (“reasonable” widths under service loads): 0.016 in (0.41 mm) dry air, 0.012 in (0.30 mm) humid air/soil, 0.007 in (0.18 mm) deicing chemicals, 0.006 in (0.15 mm) seawater, 0.004 in (0.10 mm) water-retaining structures. In US units the app compares against the native inch values. The “> 2 × limit” criterion is internal to the app.

Possible causes. Heuristic compatibility ranking (score out of 100, not a probability), based on ACI 224.1R-07 and Concrete Society TR22 (4th ed., 2010). It is not a diagnosis.

5. Strength gain

EN 1992-1-1 (3.2):

$$f_{cm}(t) = \beta_{cc}(t) \cdot f_{cm} \quad \text{with} \quad \beta_{cc}(t) = \exp\left\{s\left[1 - \left(\frac{28}{t}\right)^{1/2}\right]\right\}$$

s = 0.20 (class R cements), 0.25 (class N), 0.38 (class S) — EN 1992-1-1 §3.1.2(6).

ACI 209R-92: $f(t) = \frac{t}{a + \beta t} f_{28}$, with a = 4.0 and β = 0.85 (moist-cured, type I cement). With ACI, the default 28-day target is f'cr (ACI 318-19, no data).

Equivalent age (maturity, ASTM C1074). Nurse-Saul and Arrhenius, and EN 1992-1-1 Annex B (B.10):

$$t_e = \sum_i \frac{T_i - T_0}{T_r - T_0} \Delta t_i \quad ; \quad t_e = \sum_i e^{-\frac{E_a}{R}\left(\frac{1}{273.15+T_i} - \frac{1}{273.15+T_r}\right)} \Delta t_i \quad ; \quad t_T = \sum_i e^{-\left(\frac{4000}{273+T_i} - 13.65\right)} \Delta t_i$$

T_r = 20 °C (68 °F); default values E_a = 41.5 kJ/mol and T_0 = 0 °C, to be determined by testing the mix. Trapezoidal integration, last temperature held until t. Calibration on test results adjusts the scale (1 test) or the scale and the shape parameter (≥ 2 tests) by least squares.

6. Rebar, bar shapes and mesh

Mass per metre: $m = \rho \frac{\pi d^2}{4}$ with ρ = 7850 kg/m³ (EN 10080), i.e. ≈ 0.006 17 d² kg/m (d in mm). Mesh: mass per area = sum of both directions, $A_s \,/\, s = \frac{\pi \varphi^2}{4} \cdot \frac{1000}{s}$ mm²/m. Excludes laps, offcuts and tying wire.

Bar shapes (BS 8666:2020). Shape codes drawn: 00 (straight), 11 (L), 21 (U), 51 (closed link), overall dimensions A, B, C per the standard's convention. ISO 4066 has been withdrawn. Cutting length and minimum bending radius are **not** computed: they are determined with the formulas and tables of BS 8666:2020 (the 2020 revision changed some formulas from 2005).

6.1 Lapping of welded mesh — EN 1992-1-1:2004 §8.7.5 (ribbed wires)

$$f_{ctd} = \frac{\alpha_{ct} f_{ctk,0.05}}{\gamma_C} \quad (3.16) \qquad f_{bd} = 2.25 \eta_1 \eta_2 f_{ctd} \quad (8.2) \qquad l_{b,rqd} = \frac{\varphi}{4} \frac{\sigma_{sd}}{f_{bd}} \quad (8.3)$$

$$l_0 = \alpha_1 \alpha_2 \alpha_3 \alpha_5 \alpha_6 l_{b,rqd} \geq l_{0,\min} = \max\left\{0.3 \alpha_6 l_{b,rqd};\, 15\varphi;\, 200\text{mm}\right\} \quad (8.10),\ (8.11)$$

- $f_{ctk,0.05}$: Table 3.1, capped at the C60/75 value for bond (§8.4.2(2)); η_1 = 1.0 (good bond) or 0.7; η_2 = 1.0 for φ ≤ 32 mm.
- $\sigma_{sd} = f_{yd} = f_{yk}/\gamma_S$ by default (conservative). Twin wires: $\varphi_n = \varphi\sqrt{2}$ (§8.4.3(4)).
- α_2 = 1 − 0.15 (c_d − φ)/φ bounded to [0.7; 1.0] (Table 8.2); α_5 = 1 − 0.04 p bounded to [0.7; 1.0]; $\alpha_2\alpha_3\alpha_5 \geq 0.7$ (8.5); α_3 = 1.0 (intermeshed: §8.7.5.1(3); layered: conservative app choice); $\alpha_6 = (\rho_1/25)^{0.5}$ bounded to [1.0; 1.5] (Table 8.3).

- Layered sheets: 100 % may be lapped if $(A_s/s)_{prov} \le 1200$ mm²/m, otherwise 60 % (§8.7.5.1(6)); joints staggered by ≥ 1.3 l_0; no additional transverse reinforcement (§8.7.5.1(7)); stress > 80 % at the lap: apply §8.7.5.1(5); fatigue: intermeshing required (§8.7.5.1(2)).
- Distribution reinforcement (§8.7.5.2, Table 8.4): φ ≤ 6 mm: ≥ 150 mm and 1 pitch; 6 < φ ≤ 8.5: ≥ 250 mm and 2 pitches; 8.5 < φ ≤ 12: ≥ 350 mm and 2 pitches.

$\gamma_C = 1.5$, $\gamma_S = 1.15$, $\alpha_{ct} = 1.0$ are the **recommended** values (Table 2.1N, §3.1.6); the National Annex prevails. The UK National Annex is not built in. Plain wires are not covered by §8.7.5. No equivalent ACI 318 rule has been verified: mesh lapping is computed to Eurocode 2 only.

Example (checkable): C30/37, Ø6 @ 150 mm, good bond, B500: $f_{ctd} = 2.0/1.5 = 1.333$ MPa; $f_{bd} = 3.00$ MPa; $l_{b,rqd} = 6/4 \times 434.8/3.00 = 217$ mm; $c_d = 30$ mm ⇒ $\alpha_2 = 0.7$; $\rho_1 = 100$ % ⇒ $\alpha_6 = 1.5$; $l_0 = 0.7 \times 1.5 \times 217 = 228$ mm (≥ 200 mm). With $\rho_1 = 25$ %: $l_0 = 200$ mm (minimum).

7. Concrete volumes

Theoretical geometric volumes (duct bank, manhole, U-ditch), excluding waste, over-thickness, chamfers and openings not entered. Duct opening: $Ø_{open} = Ø_{ext}$ + 2 × clearance; axis = IL − t_{wall} + $Ø_{ext}/2$. Self-weight: 25 kN/m³ (EN 1991-1-1 Table A.1, normal-weight reinforced concrete), mass = weight/g with g = 9.81 m/s². In US units: 150 lb/ft³, the usual US design value — *I don't know* a precise normative table to cite for this value.

8. Mortar

ASTM C270, proportion specification (Table 1, cement–lime by volume): lime per part of cement M 0.25; S 0.25–0.5; N 0.5–1.25; O 1.25–2.5; sand 2.25 to 3 times the sum of binder volumes. Reference bulk densities 94 / 40 / 80 lb/ft³ (cement / hydrated lime / dry sand). Minimum 28-day laboratory strengths (property specification, Table 2): 2500, 1800, 750 and 350 psi (17.2; 12.4; 5.2; 2.4 MPa) for M, S, N, O. Absolute-volume balance, with air and a starting water/binder ratio (to be adjusted for consistency). Cement–sand: free ratio; the EN 998-2 class must be demonstrated by testing (EN 1015-11).

9. Site report

9.1 Shifts

Default hours: day 06:30–17:30, night 17:30–04:30 (next day); the two hours 04:30–06:30 are covered by no shift. Half-open bounds [start, end): at exactly 17:30 you are in the night shift. The night shift is dated on the day it starts. Hours are frozen in each report when it is created. The window is computed with the calendar of the site time zone (manual setting, prefilled from the location when Apple Maps provides it): a shift spanning a daylight-saving change really lasts 10 or 12 h instead of 11. One report per shift: the app flags a probable duplicate (same place, day, shift and trade).

9.2 Heat Index

National Weather Service formula (Rothfusz), T in °F and RH in %:

$$\mathrm{HI} = -42.379 + 2.049\,015\,23T + 10.143\,331\,27R - 0.224\,755\,41TR - 6.837\,83\cdot 10^{-3}T^2 - 5.$$

Steadman's simple formula, 0.5 [T + 61 + 1.2 (T − 68) + 0.094 R], is used as long as its average with T stays below 80 °F; NWS adjustments for R < 13 % (80–112 °F) and R > 85 % (80–87 °F). NWS categories: caution 80–90 °F, extreme caution 91–103 °F, danger 104–124 °F, extreme danger ≥ 125 °F. Shade value: full sunshine can add up to 15 °F (NWS). It is neither a WBGT nor a work-stoppage threshold.

Per shift: Apple Weather hourly records over the window (± 1 h to interpolate the bounds). The Heat Index is computed after linear interpolation of T and RH at the bounds. Maximum and peak time over [start, end); time-weighted trapezoidal mean over [start, end]:

$$\bar{\mathrm{HI}} = \frac{1}{t_n - t_0}\sum_{k=1}^{n}\frac{\mathrm{HI}_{k-1} + \mathrm{HI}_k}{2}\left(t_k - t_{k-1}\right)$$

The report states the number of observed and forecast records and flags a partially covered window. Rain: total prorated to the hours overlapping the window. Reports created before shifts were added show “shift not set” and the day's weather 6 am–6 pm.

10. Sharing and import (.civilsuite) — Premium

- Format: “CVSX” signature, uncompressed header (title, declared author, list of items), binary property list payload compressed with LZFSE, SHA-256 check digest. A unique identifier per object prevents duplicates.
- The **author is declared** freely: it is neither a signature nor an authentication. The provenance chain (successive authors, dates) is kept.
- “Remove GPS coordinates” option (on by default). Photos are always re-encoded: only the capture date and orientation are kept (camera model and other metadata removed).
- Import: the file is treated as untrusted input (size cap, digest, structure, known modules, finite values). Duplicate policy: copy, replace if newer, skip. A free user sees

a read-only preview (title, declared author, calculation types) before the Premium screen.
- Values are in SI: the recipient regenerates the PDFs in their own language and units.

11. References and verification status

Reference	Use	Status
EN 1992-1-1:2004 (Eurocode 2) — §3.1, Tables 2.1N, 3.1, (3.2), (3.16), §8.4, §8.7.3, §8.7.5, Table 7.1N, Tables 8.2–8.4, Annex B	Strengths, bond, laps, cracking, maturity	Checked against the standard text (BSI EN version supplied by the user)
EN 206 — Annex F, 1.48 σ criterion	Exposure, margin	Recommended values; the National Annex prevails
ACI 318-19 §26.4.3.1, Table 19.3.2.1	f′cr, minimum f′c	Checked on secondary sources (NRMCA articles / technical documents); ACI text not consulted directly
ACI 211.1-91 (R2009), Tables 6.3.1, 6.3.3, 6.3.4(a), 6.3.6	ACI mix design	Tables built into the initial app version (metric appendix)
ACI 209R-92; ASTM C1074	Strength gain, maturity	Reference formulas
ACI 224R-01 Table 4.1; ACI 224.1R-07; Concrete Society TR22 (2010)	Cracks	Table 4.1 values in inches and mm
BS 8666:2020	Shape codes	Codes and dimension letters only; length formulas not built in
EN 10080 (ρ = 7850 kg/m³); EN 1991-1-1 Table A.1 (25 kN/m³)	Masses, self-weight	Usual values
ASTM C270	Mortar	Proportions and strengths (Tables 1 and 2)
NWS — The Heat Index Equation	Heat Index	wpc.ncep.noaa.gov
NIST SP 811, Appendix B	Conversions	nist.gov
US value 150 lb/ft³	US self-weight	I don't know (no normative table verified)

Weather data: Apple Weather. Standards are protected documents: only their references and the necessary formulas are reproduced.